

6 Sigma Assignment

1. Define

- 1) Define Issue
- 2) NG Rate & PPM

2. Measurement

- 1) Gage R&R outline
- 2) Type 1 Study
- 3) Type 2 Study
- 4) Process Capability Review
- 5) Conclusion

3. Analyze

- 1) Find the Root Cause

4. Improvement

- 1) Improvement activity
- 2) Check Process Capability

5. Control

2021.11.04

6 Sigma GB assignment
IVI PD – Nguyen Sy Tuan

Six Sigma Theme Registration

LGE Internal Use Only

Team	IVI LD
T/L	Lee Sangman
Owner	Nguyen Sy Tuan

Theme	Bluetooth Distortion NTF of JLR P-IVI				Period	2021.Nov.04~2022.Feb.04			
Improve item	Target (Stretch Goal)				Idea (How to improve)				
	KPI	Current	World Best	Target					
- Reduce NTF rate of Bluetooth distortion to 0 ppm	Total defect W40 (ppm)	11628 (ppm)		0 ppm (100%↓)	- Separate ground for each equipment - Increase delay time for measurement				
Current Issue	Planning (Fast & Strong)				Organization				
1. Measurement data is near USL 2. Defect rate is very high 3. Issue occurs in the long time	D 04/11 - Define issue - Define defect rate	M 11/11 - Gage R&R evaluate equipment	A 18/11 - Test sample with using different spec - Discuss with HQ to improve this issue	I 26/11 - Separate ground for each equipment - Increase delay time for measurement	C 10/12 - Control defect rate of NTF issue through GMES system	Name	Dept	Full Time	Part Time
						N.S.Tuan	LD		O
						N.N.An	PE	O	
						N.D.Quang	PM	O	
						P.P.Thao	MFG	O	
						Location			
						LG Electronics Viet Nam Hai Phong			

1.1 Define Issue

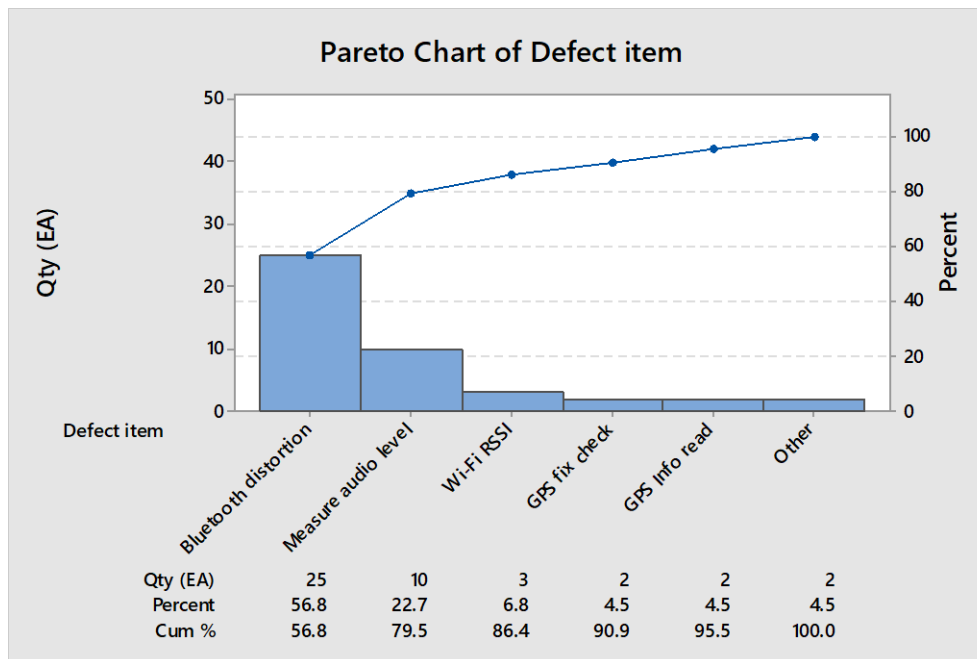
❑ Issue information

- Issue: Bluetooth distortion test fail
- Model: JLR P-IVI
- Defect type: 25 / 2150 (11628 ppm)

❑ Reason of choosing

- This issue occurs with high defect rate
- I choose this item because high NTF occurs frequently, UPH will be reduced

1.2 Defect rate



Defect symptom	Qty (EA)	%
Bluetooth distortion	25	56.8%
Measure audio level	10	4.5%
GPS fix check	3	22.7%
Wi-Fi RSSI	2	4.5%
Sleep current	2	6.8%
Other	2	4.5%
Total	44	100%

Bluetooth distortion is worst defect of this part
 → I select this defect to make my 6sigma assignment

2. Measurement

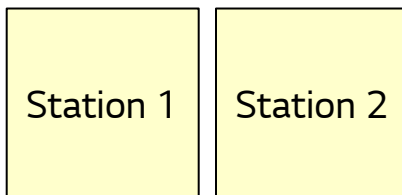
2.1 Gage R&R outline

- Name: Bluetooth distortion
- Part: JLR P-IVI
- Spec: 0 ~ 9 Vrms
- Method: measure Bluetooth distortion when play mp3 file
- Equipment: Audio Video Multi Tester (AVMT)
- Resolution: 0.000001Vrms

Audio Video Multi Tester



Bluetooth station



05_JLR_BT_22042021							
PASS							
Progress		100%					
NO.	TEST NAME	RESULT	PAR	MIN	MAX	MEASURE	LAPSE
27	<TRY.START>.	PASS				3	0.07
28	JLR.BT.RSSI.	PASS		174	236	205	0.09
29	<TRY.END>.	PASS					0.06
30	MP3.PLAY.START.1KHz.mp3	PASS	1KHz.mp3				0.08
31	<TRY.START>.	PASS				3	0.06
32	JLR.BT.LOOP.BACK.TEST.	PASS					0.08
33	AVMT.MEAS.DISTN.MAIN	PASS	MAIN	0	9	0.72867000,0.72972000	0.75
34	JLR.BT.LOOP.BACK.STOP.	PASS					0.09
35	<TRY.END>.	PASS					0.06
36	MP3.PLAY.STOP.	PASS					0.07
37	JLR.BT.CLEAR(UNPAIR).	PASS					0.14
38	BT.CANCEL.	PASS				DISCONNECT	0.10
39	JLR.BT.DISCOVERABLE.DISABLE.	PASS					0.07
40	JLR.AUDIO.STOP.	PASS				1	0.38
41	MES.STEP.COMPLETE.	PASS				SUCCESS	0.41
42	DIO.RELAY.ALL.CLEAR.	PASS					0.09

2. Measurement

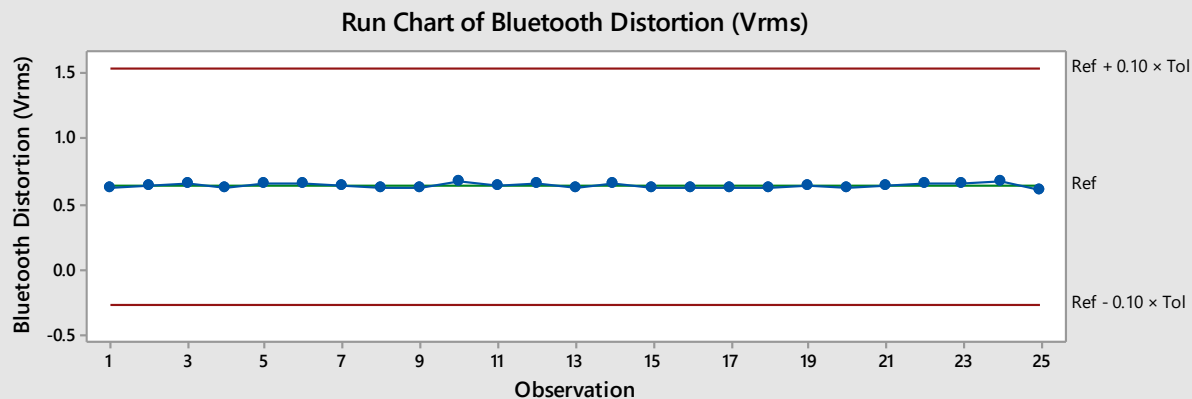
2.2 Type 1 Study (Station 1)

No	Bluetooth Distortion (Vrms)	No	Bluetooth Distortion (Vrms)
1	0.64436	26	0.65231
2	0.64901	27	0.63750
3	0.67625	28	0.64185
4	0.63142	29	0.66580
5	0.66855	30	0.65255
6	0.67168		
7	0.65221		
8	0.6352		
9	0.63904		
10	0.67939		
11	0.65681		
12	0.67688		
13	0.63755		
14	0.67726		
15	0.63269		
16	0.63062		
17	0.63636		
18	0.63316		
19	0.66101		
20	0.64196		
21	0.65418		
22	0.66586		
23	0.67449		
24	0.67951		
25	0.62231		

Type 1 Gage Study for Bluetooth Distortion (Vrms)

Gage name:
Date of study:

Reported by:
Tolerance: 9
Misc:



Basic Statistics

Reference	0.653
Mean	0.65311
StDev	0.018654
6 × StDev (SV)	0.111923
Tolerance (Tol)	9
Resolution	0.000001 < 5% of Tol

Bias

Bias	0.00011
T	0.029592
PValue	0.977
(Test Bias = 0)	

Capability

Cg	16.08
Cgk	16.08
%Var(Repeatability)	1.24%
%Var(Repeatability and Bias)	1.24%

Capability

- Cg = 16.08 (Standard > 1.33) → OK
- Cgk = 16.08 (Standard > 1.33) → OK
- %Var(Repeatability) = 1.24% (Standard < 10%) → OK
- %Var(Repeatability and Bias) = 1.24% (Standard < 10%) → OK
- Resolution < 5% of Tolerance → OK

Conclusion: The AVMT1 can be used to measure Bluetooth distortion

2.2 Type 1 Study (Station 1)

- NTF issue only happen at station 2, station 1 is OK, so I get reference value base on data in station 1
- Reference value: 0.654711

No	Bluetooth Distortion (Vrms)
1	0.64436
2	0.64901
3	0.67625
4	0.63142
5	0.66855
6	0.67168
7	0.65221
8	0.6352
9	0.63904
10	0.67939

2.2 Type 1 Study (Station 2)

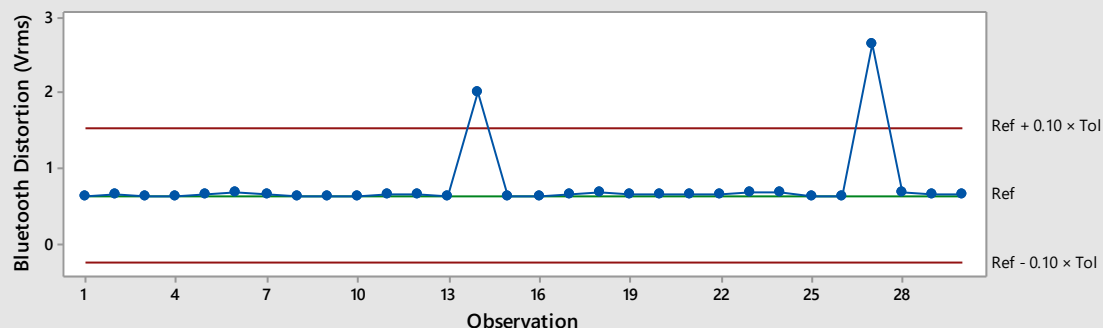
No	Bluetooth Distortion (Vrms)	No	Bluetooth Distortion (Vrms)
1	0.65069	26	0.6409
2	0.66348	27	2.65301
3	0.6482	28	0.69068
4	0.64339	29	0.65667
5	0.67777	30	0.66561
6	0.68774		
7	0.66283		
8	0.64107		
9	0.64777		
10	0.63812		
11	0.67994		
12	0.68159		
13	0.6356		
14	2.0102		
15	0.6348		
16	0.64297		
17	0.66025		
18	0.6855		
19	0.67173		
20	0.65726		
21	0.66568		
22	0.68177		
23	0.68727		
24	0.6883		
25	0.64914		

Type 1 Gage Study for Bluetooth Distortion (Vrms)

Gage name:
Date of study:

Reported by:
Tolerance: 9
Misc:

Run Chart of Bluetooth Distortion (Vrms)



Basic Statistics

Reference	0.654711
Mean	0.77333100
StDev	0.432279530
6 × StDev (SV)	2.593677180
Tolerance (Tol)	9
Resolution	0.000001 < 5% of Tol

Bias

Bias	0.11862000
T	1.502982336
PValue	0.144
(Test Bias = 0)	

Capability

Cg	0.69
Cgk	0.60
%Var(Repeatability)	28.82%
%Var(Repeatability and Bias)	33.19%

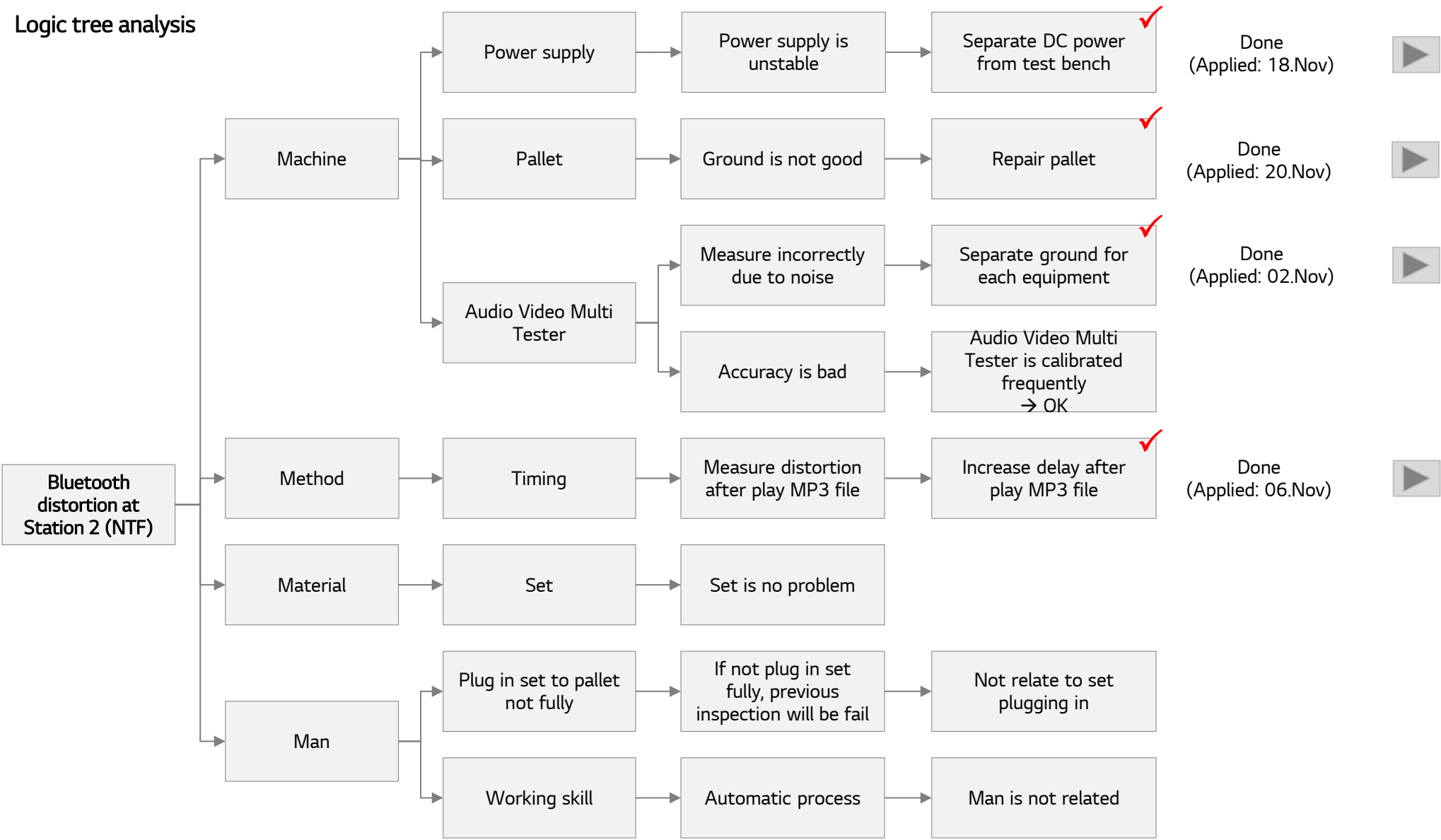
Capability

- Cg = 0.69 (Standard > 1.33) → NG
- Cgk = 0.60 (Standard > 1.33) → NG
- %Var(Repeatability) = 28.82% (Standard > 10%) → NG
- %Var(Repeatability and Bias) = 33.19% (Standard > 10%) → NG
- Resolution < 5% of Tolerance → OK

Conclusion: The AVMT2 need to be improved

3.1 Find the Root Cause

Logic tree analysis



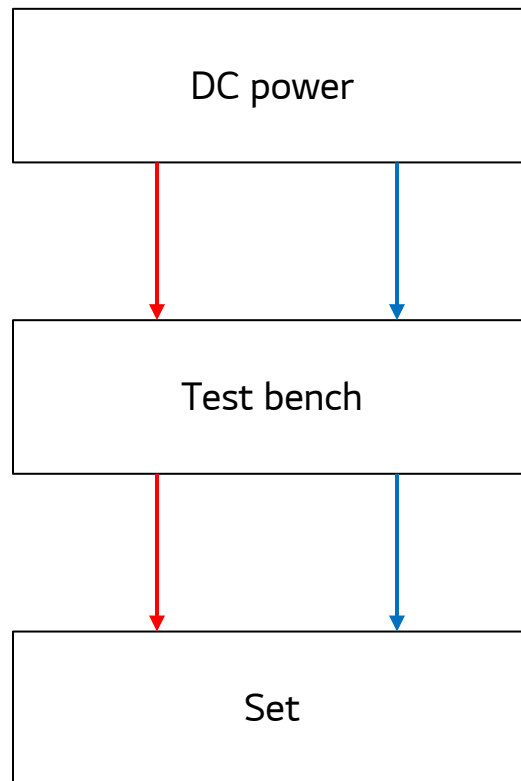
4. Improvement

4.1 Separate DC power from test bench

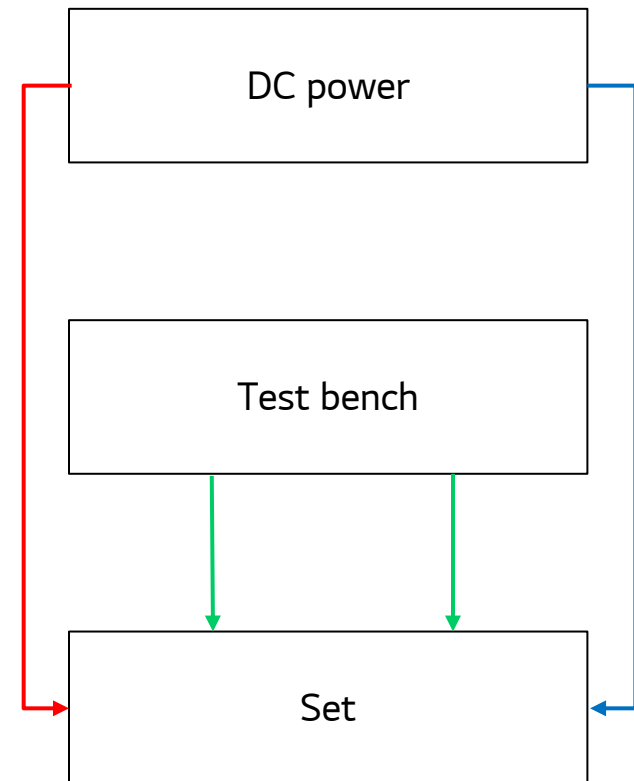
- Test bench includes SMPS (Switching Mode Power Supply). The power quality of SMPS is not good (high ripple), it can make noise and affect to measurement data.

→ Separate power supply from test bench to improve power quality.

As is



To be



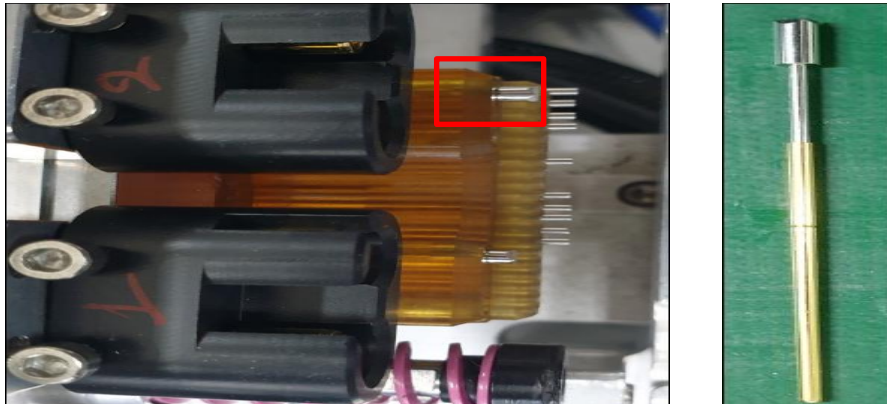
4. Improvement

4.2 Repair pallet

- Bluetooth distortion is measured through pogo pin from Set to AVMT. So, if the ground connection is not good, it can make noise and affect to measurement data

→ Add more ball bearing for ground pogo pin to increase electrical conductivity

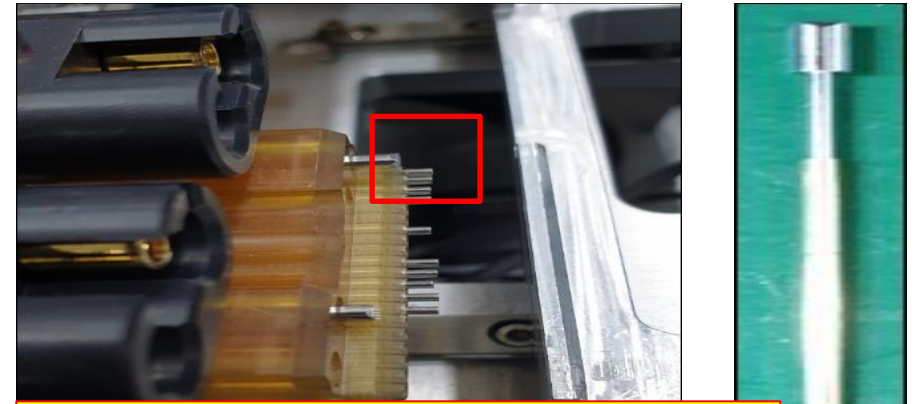
Before



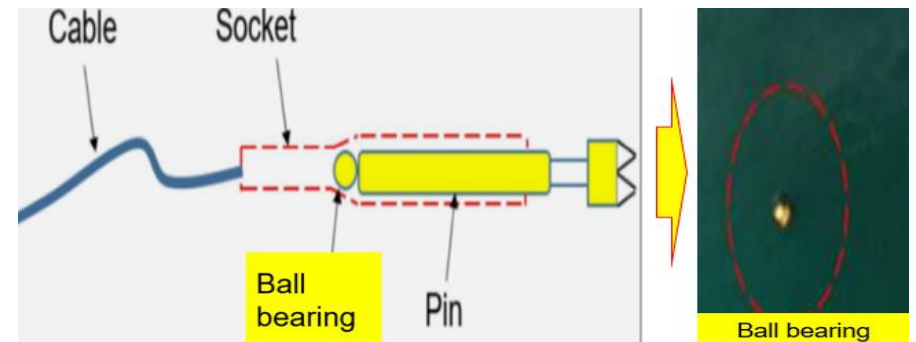
Case Ground Pogo pin no use ball bearing



After



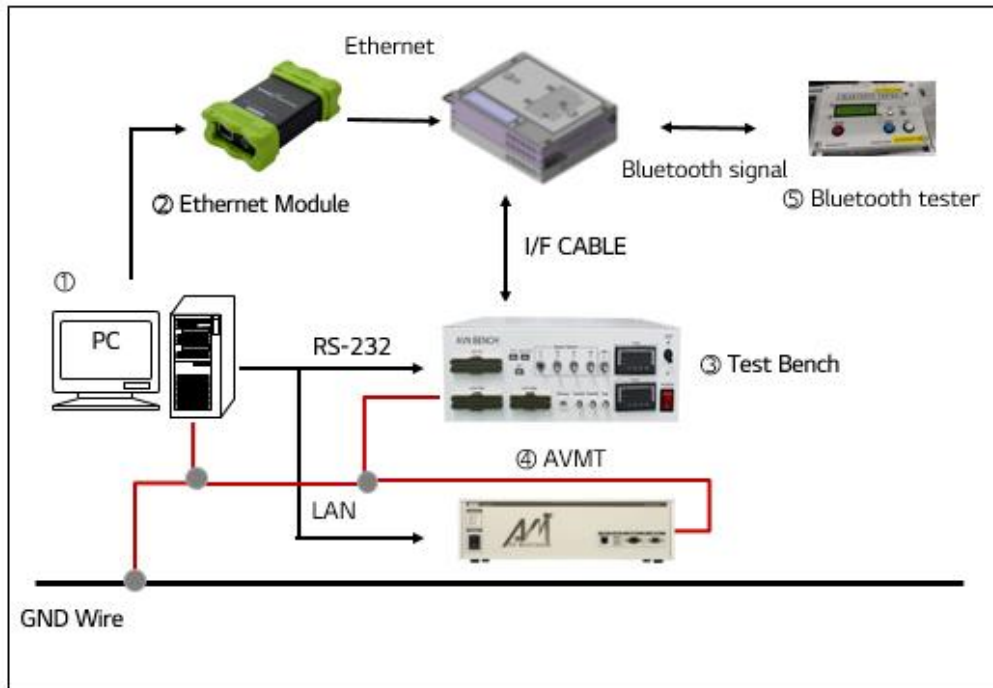
Adding ball bearing for Ground Pogo Pin



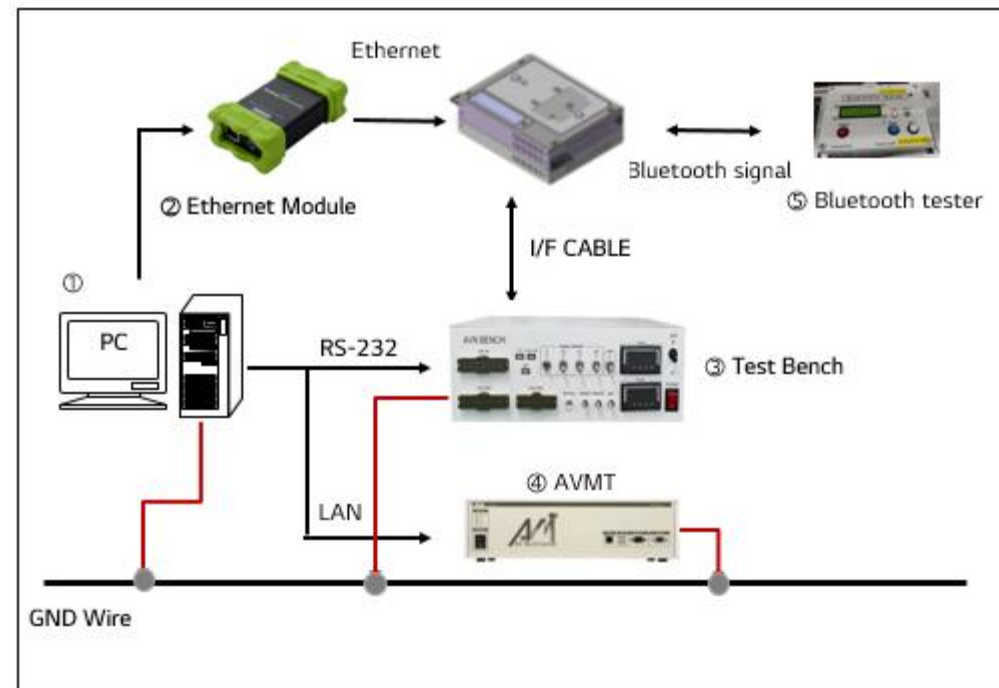
4. Improvement

4.3 Separate ground for each equipment

- Before, ground of each equipment is connected with each other, so when measuring Bluetooth distortion data, power noise of other equipment can affect to AVMT, it makes data incorrect
- Separate ground of all equipment.



Use common GND Wire



Separate GND Wire

4.4 Increase delay after play MP3 file

- Bluetooth distortion is measured after Set play MP3 file completely. The measured data is not a stable if Set not yet play file.
- Add 1.5s delay for playing MP3 file, Bluetooth signal will be stable, so AVMT will measure data correctly.

Before

31	<TRY.START>	3		RUN	CONTINUE	0		NONE		
32	JLR.BT.RSSI		R_BT_RS	RUN	STOP	0		NUMBER	174	236
33	<TRY.END>			RUN	CONTINUE	0		NONE		
34	MP3.PLAY.START			RUN	CONTINUE	0	1KHz.mp3	NONE		
35	<TRY.START>	5		RUN	CONTINUE	0		NONE		
36	JLR.BT.CALL.ACCEPT			SKIP	CONTINUE	3		NONE		
37	AVMT.SIGNAL.VOLT			SKIP	CONTINUE	0	LR,1.0	NONE		
38	JLR.BT.LOOP.BACK.TEST			RUN	CONTINUE	0		NONE		
39	AVMT.MEAS.LEVEL			SKIP	CONTINUE	0	MAIN	NUMBER	0	1
40	DELAY			RUN	CONTINUE	0.5		NONE		
41	AVMT.MEAS.DISTN			RUN	CONTINUE	0	MAIN	NUMBER	0	9
42	JLR.BT.LOOP.BACK.STOP			RUN	CONTINUE	0		NONE		

After

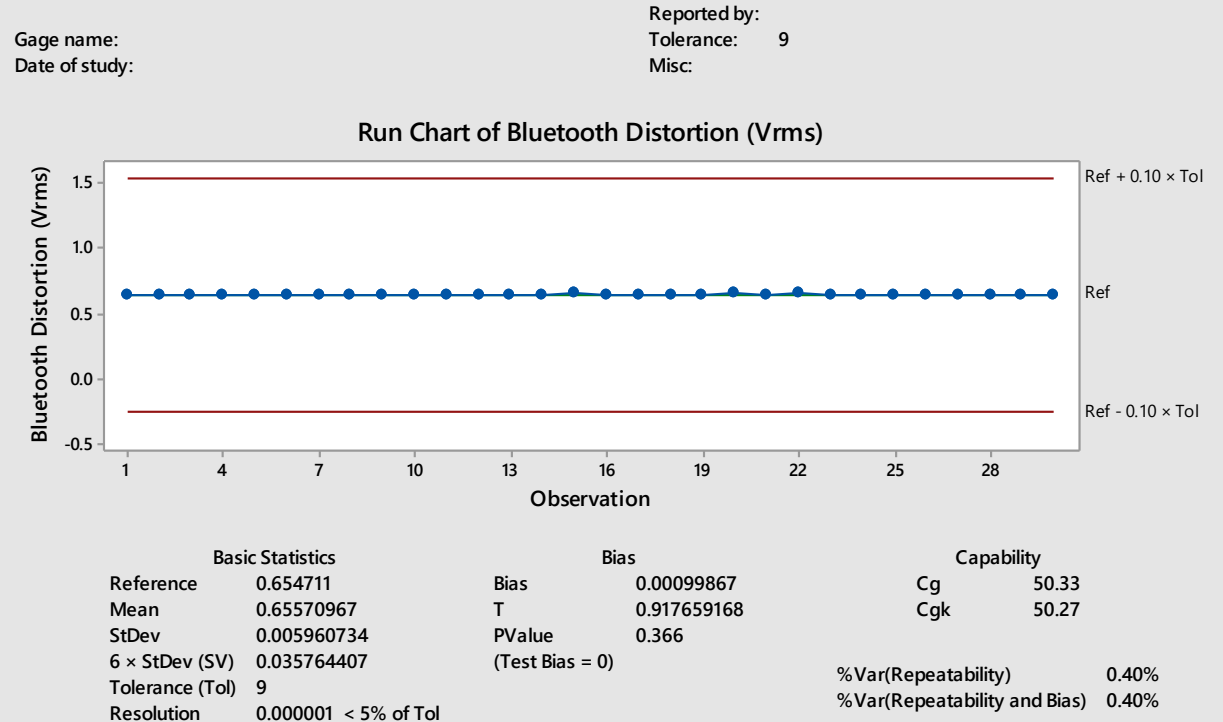
31	<TRY.START>	3		RUN	CONTINUE	0		NONE		
32	JLR.BT.RSSI		R_BT_RS	RUN	STOP	0		NUMBER	174	236
33	<TRY.END>			RUN	CONTINUE	0		NONE		
34	MP3.PLAY.START			RUN	CONTINUE	0	1KHz.mp3	NONE		
35	DELAY			RUN	CONTINUE	1.5		NONE		
36	<TRY.START>	5		RUN	CONTINUE	0		NONE		
37	JLR.BT.CALL.ACCEPT			SKIP	CONTINUE	3		NONE		
38	AVMT.SIGNAL.VOLT			SKIP	CONTINUE	0	LR,1.0	NONE		
39	JLR.BT.LOOP.BACK.TEST			RUN	CONTINUE	0		NONE		
40	AVMT.MEAS.LEVEL			SKIP	CONTINUE	0	MAIN	NUMBER	0	1
41	DELAY			RUN	CONTINUE	0.5		NONE		
42	AVMT.MEAS.DISTN			RUN	CONTINUE	0	MAIN	NUMBER	0	9
43	JLR.BT.LOOP.BACK.STOP			RUN	CONTINUE	0		NONE		

4. Improvement

4.5 Type 1 Study at station 2

No	Bluetooth Distortion (Vrms)	No	Bluetooth Distortion (Vrms)
1	0.65436	26	0.65069
2	0.64901	27	0.65348
3	0.65625	28	0.6582
4	0.65142	29	0.65339
5	0.64855	30	0.65777
6	0.65168		
7	0.65221		
8	0.6552		
9	0.64904		
10	0.64939		
11	0.65681		
12	0.65688		
13	0.65755		
14	0.65726		
15	0.67269		
16	0.65062		
17	0.65636		
18	0.65316		
19	0.66101		
20	0.67196		
21	0.65418		
22	0.66586		
23	0.65449		
24	0.65951		
25	0.65231		

Type 1 Gage Study for Bluetooth Distortion (Vrms)



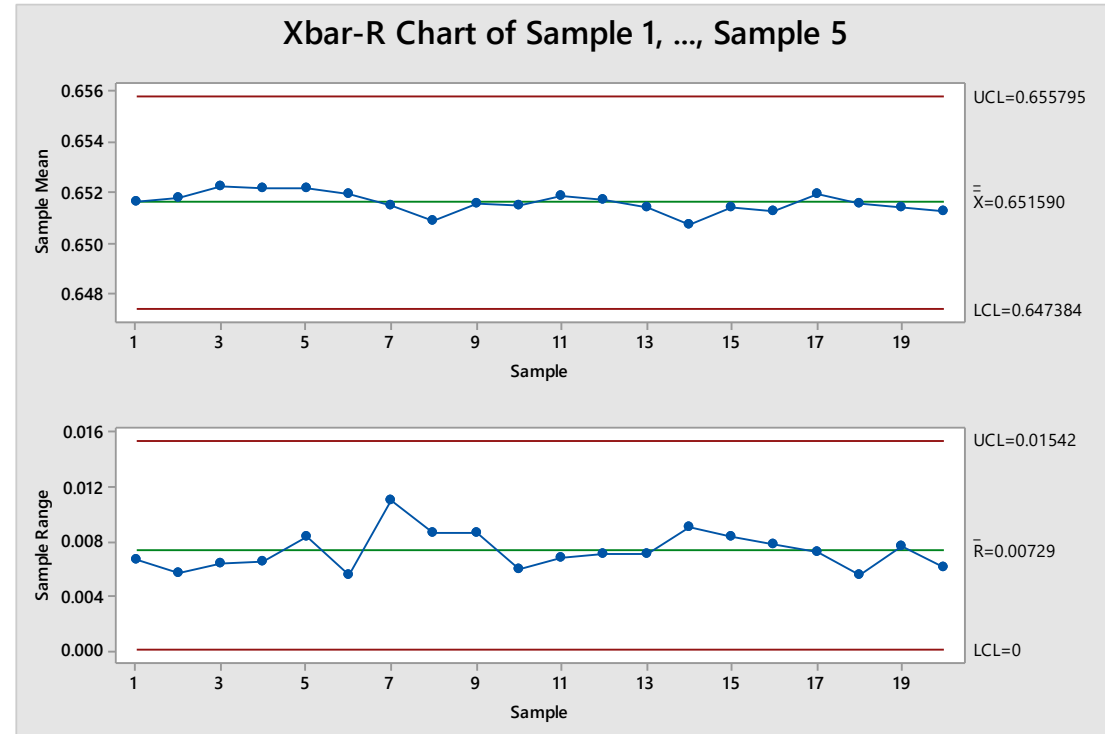
Capability

- Cg = 50.33 (Standard > 1.33) → OK
- Cgk = 50.27 (Standard > 1.33) → OK
- %Var(Repeatability) = 0.40% (Standard < 10%) → OK
- %Var(Repeatability and Bias) = 0.40% (Standard < 10%) → OK
- Resolution < 5% of Tolerance → OK

Conclusion: The AVMT2 can be used to measure Bluetooth distortion

4.6 Xbar-R Chart at station 2

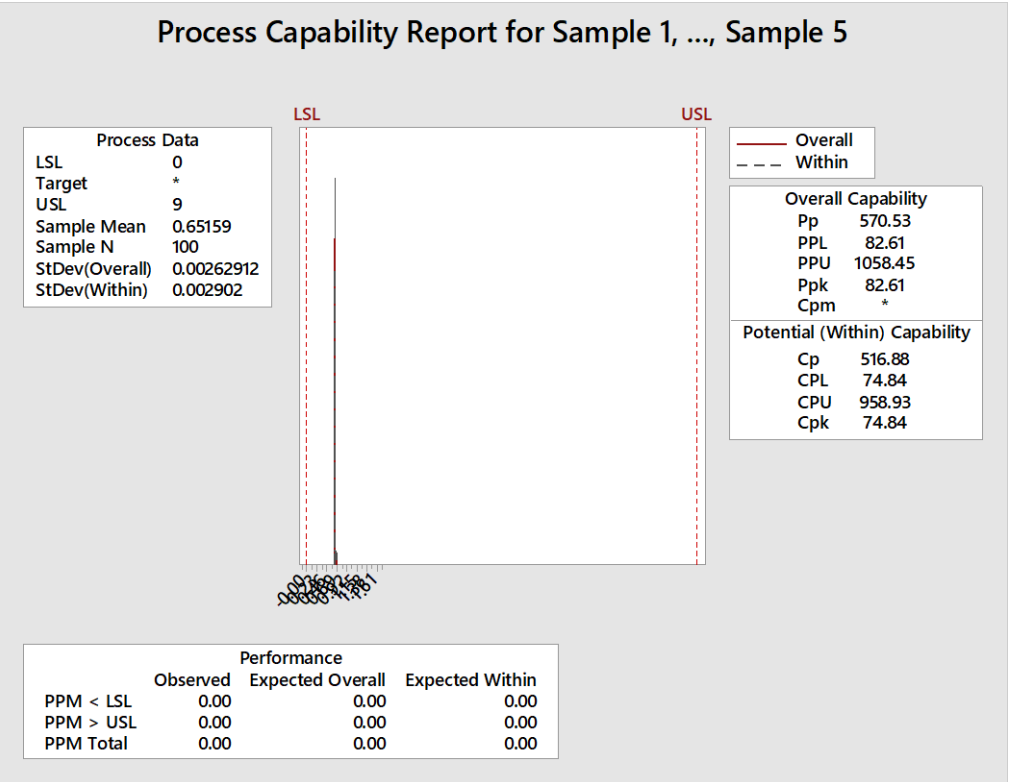
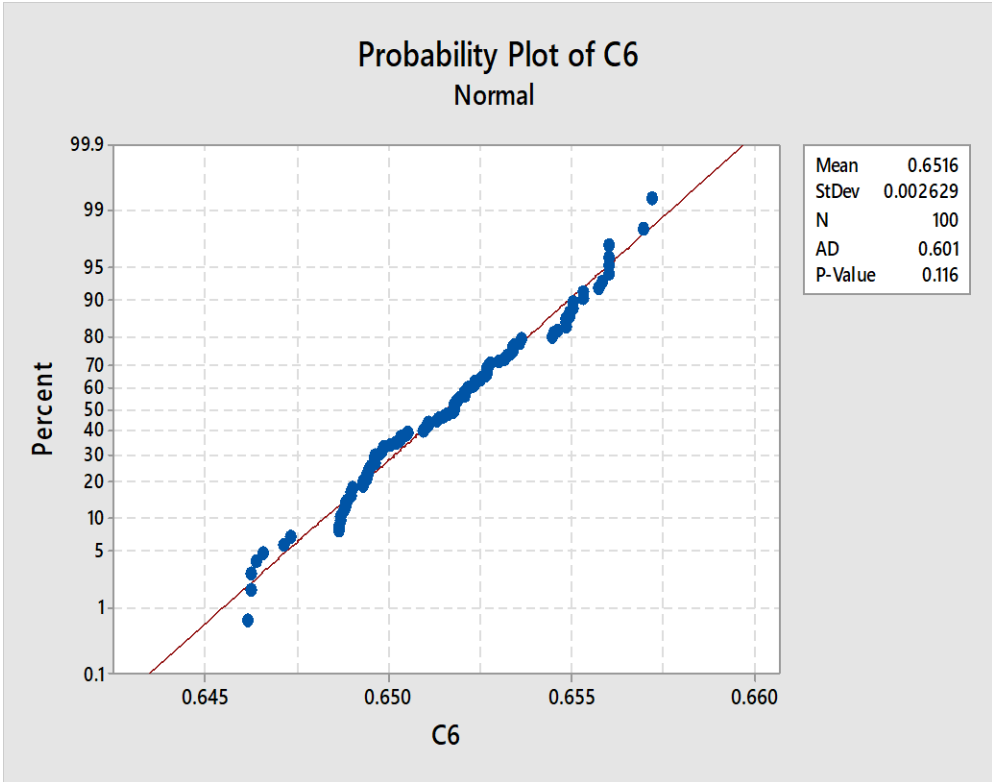
Date	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5
6-Sep	0.65531	0.65182	0.64868	0.64963	0.65270
9-Sep	0.65448	0.65279	0.64951	0.64880	0.65325
10-Sep	0.65573	0.65269	0.64941	0.64965	0.65362
11-Sep	0.65586	0.65256	0.64944	0.64939	0.65357
12-Sep	0.65698	0.65136	0.64868	0.65032	0.65344
13-Sep	0.65485	0.65272	0.64946	0.64933	0.65318
15-Sep	0.65719	0.65184	0.64615	0.65048	0.65178
16-Sep	0.65486	0.65162	0.64624	0.64983	0.65180
17-Sep	0.65506	0.65237	0.64638	0.65052	0.65333
18-Sep	0.65464	0.65104	0.64863	0.64986	0.65339
19-Sep	0.65604	0.65152	0.64929	0.64977	0.65269
20-Sep	0.65601	0.65236	0.64888	0.64956	0.65179
22-Sep	0.65601	0.65110	0.64901	0.64898	0.65207
23-Sep	0.65533	0.65135	0.64625	0.64864	0.65220
24-Sep	0.65494	0.65267	0.64659	0.65018	0.65249
25-Sep	0.65485	0.65194	0.64714	0.65029	0.65189
26-Sep	0.65603	0.65208	0.64886	0.65033	0.65231
27-Sep	0.65452	0.65176	0.64900	0.64964	0.65302
29-Sep	0.65503	0.65111	0.64735	0.65004	0.65341
30-Sep	0.65495	0.65095	0.64882	0.64965	0.65207



■ Conclusion:

- No abnormality can be found in Xbar control chart and R control chart
- Therefore stability is Pass

4.7 Process Capability at station 2



P-Value = 0.116 > 0.05 → Pass

Cpk = 74.84 (Standard ≥ 1.67) → Pass

Team: I.V.I L.D

Name: Nguyễn Sỹ Tuấn

No cheating! No discussion!
 ✱ Calculate up to 2 decimal places.

80

1. Calculate Z-value and make a decision? (10 Point)

Mr. Cong took college exams twice last year and this year.
 Last year he got 280 points, this year he got 300 points.
 Which year did he do better on the exam?

(Last year's test: overall average 250 points, standard deviation 10,
 This year's test: overall average 270 points, standard deviation 20)

Last year $Z = \frac{280 - 250}{10} = 3$ $Z_{\text{last year}} > Z_{\text{this year}} \rightarrow \text{Last year he did better.}$
 This year $Z = \frac{300 - 270}{20} = 1.5$

2. Write down full name? (10 Point)

1) VOC

voice of customer

2) VOB

voice of business

3) 5M1E

man, machine, material, method, measurement, environment

4) PFMEA

process failure mode & effect analysis

3. Calculate X bar, σ , Median? (12 Point, 4 Point each question)

2, 1, 6, 4, 5, 3, 5, 10

$\bar{X} = \frac{2+1+6+4+5+3+5+10}{8} = 4.50$; Median = 4.5 ; $\sigma = \frac{2+1+6+4+5+3+5+10}{8} = 2.78$

4. Input Key work in () ? (12 Point, 2 Point each question)

The Define stage is the stage in which the project that is important from the customer point of view and has the greatest (1) CTQ impact among tasks and processes is derived and defined.

Measure step is to measure the Z level of the selected (2) theme and to set the improvement goal and direction for the CTQ. In this step normally, we should do (3) gage R & R to confirm measurement system capability.

Analyze selects major factor by verifying the (4) potential factor of CTQ derived from the Measure step.

1/3